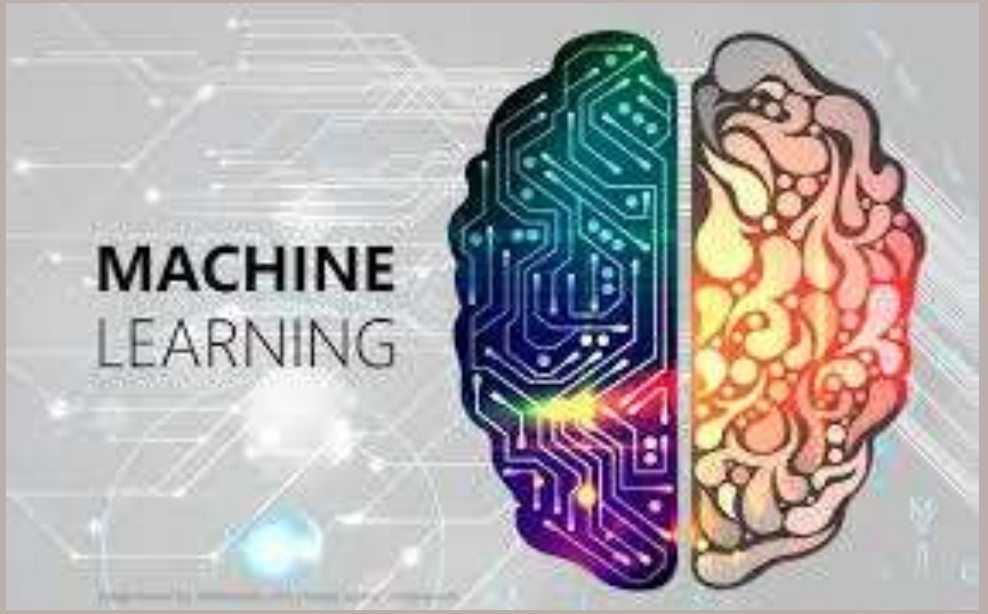
CSE 4137

Machine Learning
Module-01
Supervised ML

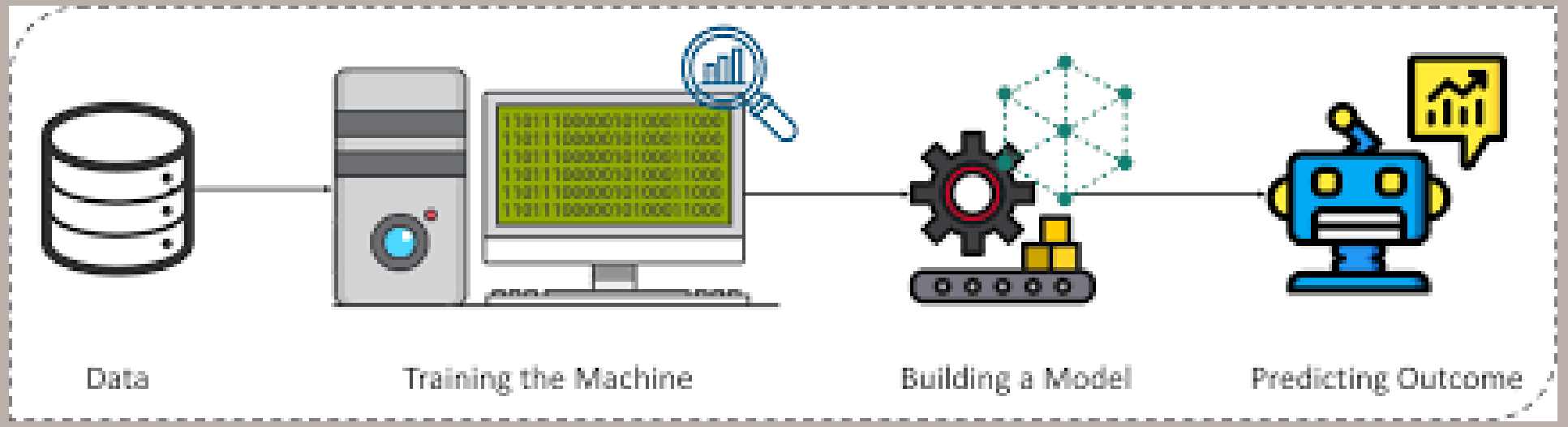


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Topics: Supervised ML

Week-02

- Basic Concepts
- Common Algorithms
- Applications



Basic Concepts: Supervised ML

Week-02, c-1

What is Supervised ML?

Supervised learning is the types of machine learning in which machines are trained using well "labelled" training data, and on basis of that data, machines predict the output. The labelled data means some input data is already tagged with the correct output.

How Supervised Learning Works?

Suppose we have a dataset of different types of shapes which includes square, rectangle, triangle, and Polygon. Now the first step is that we need to train the model for each shape.

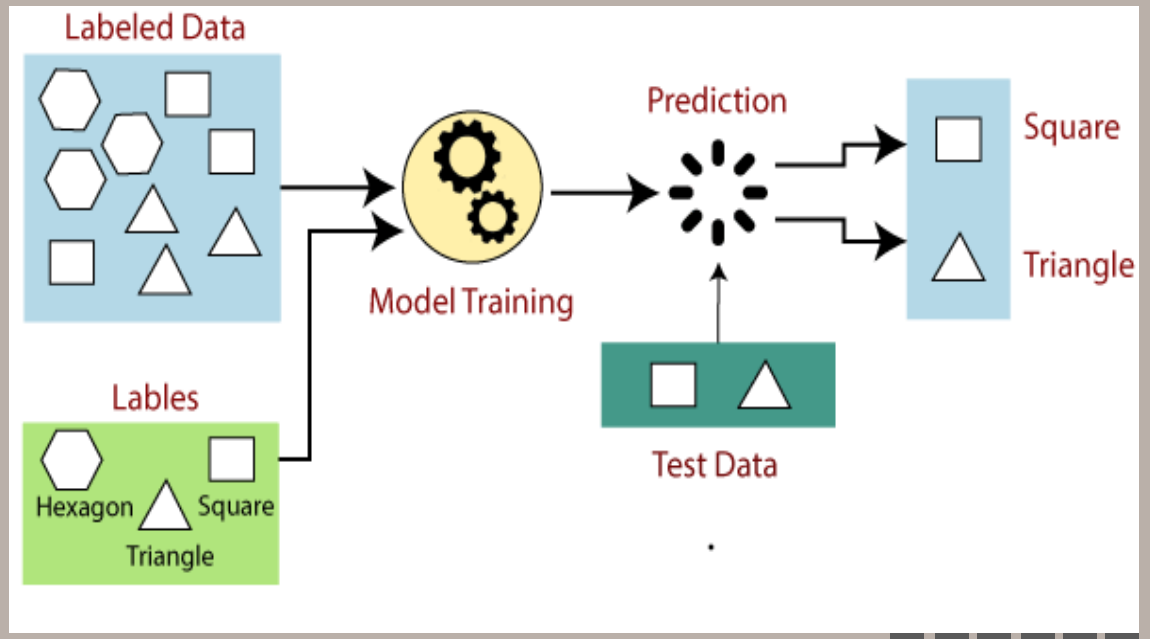
•If the given shape has four sides, and all the sides are equal, then it will be labelled as a **Square**.

•If the given shape has three sides, then it will be labelled as a **triangle**.

•If the given shape has six equal sides then it will be labelled as **hexagon**.

Now, after training, we test our model using the test set, and the task of the model is to identify the shape.

The machine is already trained on all types of shapes, and when it finds a new shape, it classifies the shape on the bases of a number of sides, and predicts the output.



Basic Concepts: Supervised ML

Week-02, c-1

Steps Involved in Supervised Learning:

•First Determine the type of training dataset
•Collect/Gather the labelled training data.
•Split the training dataset into training **dataset, test dataset, and validation dataset**.
•Determine the input features of the training dataset, which should have enough knowledge so that the model can accurately predict the output.
•Determine the suitable algorithm for the model, such as support vector machine, decision tree, etc.
•Execute the algorithm on the training dataset. Sometimes we need validation sets as the control parameters, which are the subset of training datasets.
•Evaluate the accuracy of the model by providing the test set. If the model predicts the correct output, which means our model is accurate.

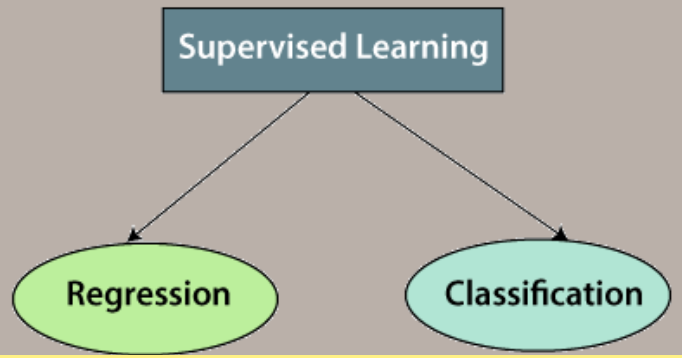
Steps in short:

```
# 0. Preprocess + EDA + Feature Selection
# 1. Extract input and output cols
# 2. Scale the values
# 3. Train test split
# 4. Train the model
# 5. Evaluate the model/model selection
# 6. Deploy the model
```

Basic Concepts: Supervised ML

Week-02, c-1

Types of Supervised ML



1. Regression algorithms are used if there is a relationship between the input variable and the output variable. It is used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc. Below are some popular Regression algorithms which come under supervised learning:

- Linear Regression
- Regression Trees
- Non-Linear Regression
- Bayesian Linear Regression
- Polynomial Regression

2. Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc. ex. Spam Filtering, Following are the popular algorithms

- Random Forest
- Decision Trees
- Logistic Regression
- Support vector Machines

Basic Concepts: Supervised ML

Week-02, c-1

Evaluating Supervised ML

For Regression

- **Mean Squared Error (MSE):** MSE measures the average squared difference between the predicted values and the actual values. Lower MSE values indicate better model performance.
- **Root Mean Squared Error (RMSE):** RMSE is the square root of MSE, representing the standard deviation of the prediction errors. Similar to MSE, lower RMSE values indicate better model performance.
- **Mean Absolute Error (MAE):** MAE measures the average absolute difference between the predicted values and the actual values. It is less sensitive to outliers compared to MSE or RMSE.
- **R-squared (Coefficient of Determination):** R-squared measures the proportion of the variance in the target variable that is explained by the model. Higher R-squared values indicate better model fit.

For Classification

- **Accuracy:** Accuracy is the percentage of predictions that the model makes correctly. It is calculated by dividing the number of correct predictions by the total number of predictions.
- **Precision:** Precision is the percentage of positive predictions that the model makes that are actually correct. It is calculated by dividing the number of true positives by the total number of positive predictions.
- **Recall:** Recall is the percentage of all positive examples that the model correctly identifies. It is calculated by dividing the number of true positives by the total number of positive examples.
- **F1 score:** The F1 score is a weighted average of precision and recall. It is calculated by taking the harmonic mean of precision and recall.
- **Confusion matrix:** A confusion matrix is a table that shows the number of predictions for each class, along with the actual class labels. It can be used to visualize the performance of the model and identify areas where the model is struggling.

Basic Concepts: Supervised ML

Week-02, c-1

Evaluating Supervised ML

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Basic Concepts: Supervised ML

Week-02, c-1

Application Supervised ML

Supervised learning can be used to solve a wide variety of problems, including:

•**Spam filtering:** Supervised learning algorithms can be trained to identify and classify spam emails based on their content, helping users avoid unwanted messages.

•**Image classification:** Supervised learning can automatically classify images into different categories, such as animals, objects, or scenes, facilitating tasks like image search, content moderation, and image-based product recommendations.

•**Medical diagnosis:** Supervised learning can assist in medical diagnosis by analyzing patient data, such as medical images, test results, and patient history, to identify patterns that suggest specific diseases or conditions.

•**Fraud detection:** Supervised learning models can analyze financial transactions and identify patterns that indicate fraudulent activity, helping financial institutions prevent fraud and protect their customers.

•**Natural language processing (NLP):** Supervised learning plays a crucial role in NLP tasks, including sentiment analysis, machine translation, and text summarization, enabling machines to understand and process human language effectively.



Example: SML



Example: Supervised ML

Week-02, c-2

Linear Regression

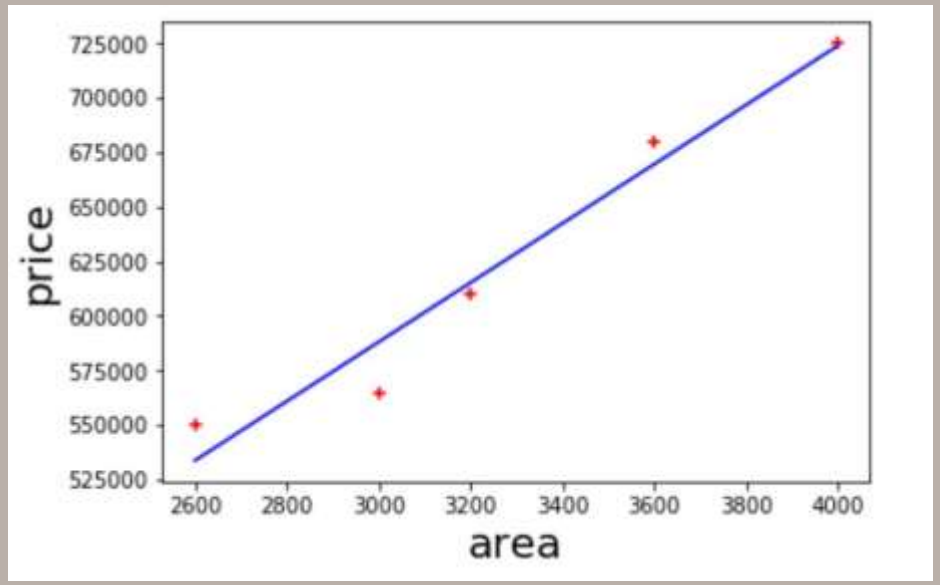
Linear regression analysis is used to predict the value of a variable based on the value of another variable. The variable you want to predict is called the dependent variable. The variable you are using to predict the other variable's value is called the independent variable.

Example: Linear Regression

Problem Statement: Given above data build a machine learning model that can predict home prices based on square feet area. Here is the sample dataset

You can represent values in above table as a scatter plot (values are shown in red markers). After that one can draw a straight line that best fits values on chart.

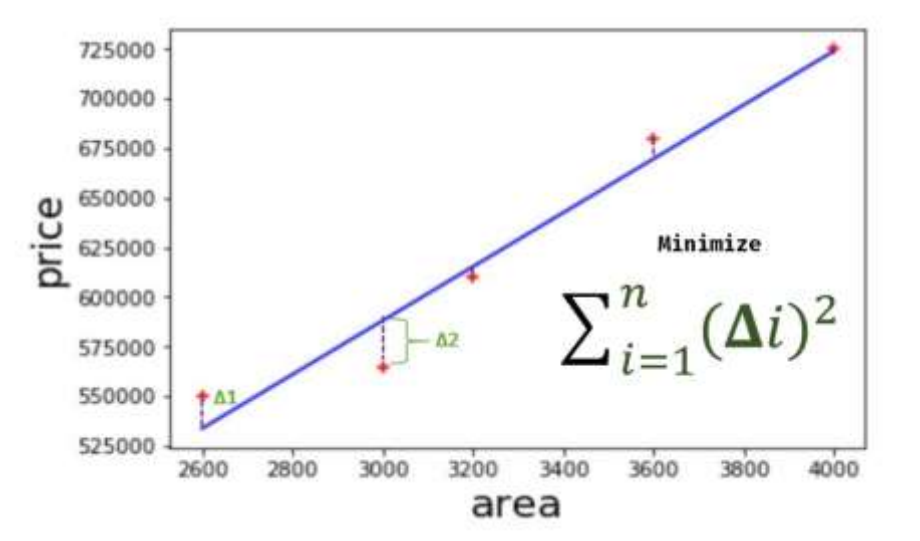
area	price
2600	550000
3000	565000
3200	610000
3600	680000
4000	725000



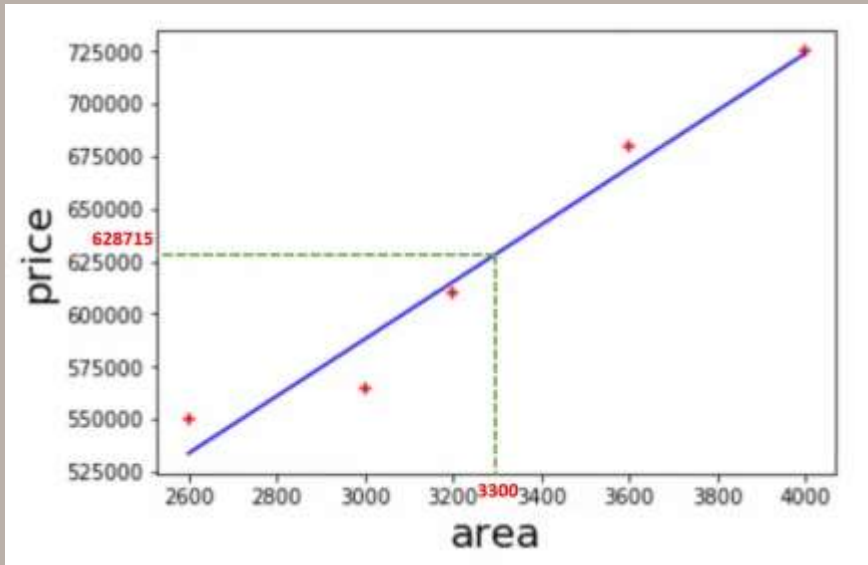
Linear Regression: SML

Week-02, c-2

Suppose you draw a line which is closes to the data points



After drawing the line you can predict the price. Remember that the area and price were not in th original dataset.

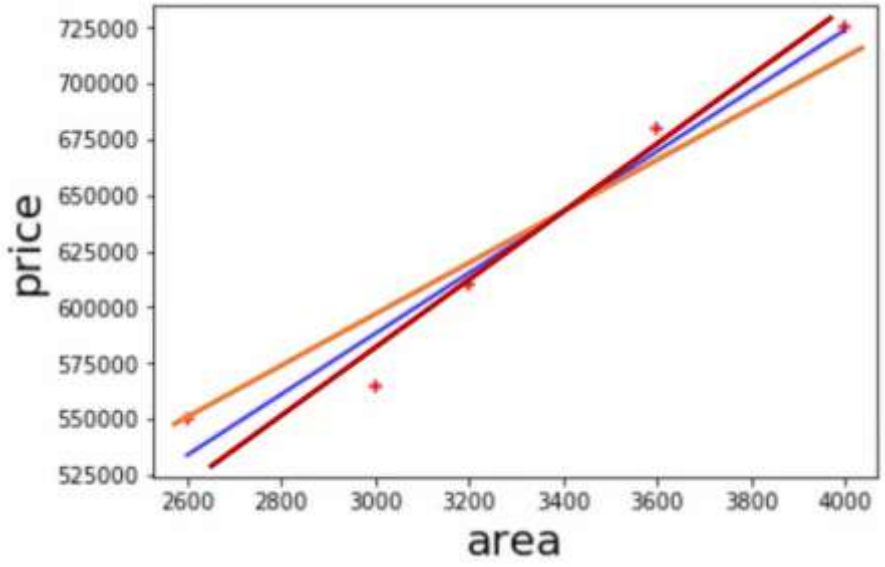


Here input area is 3300 and the predicted price is 628415

Linear Regression: SML

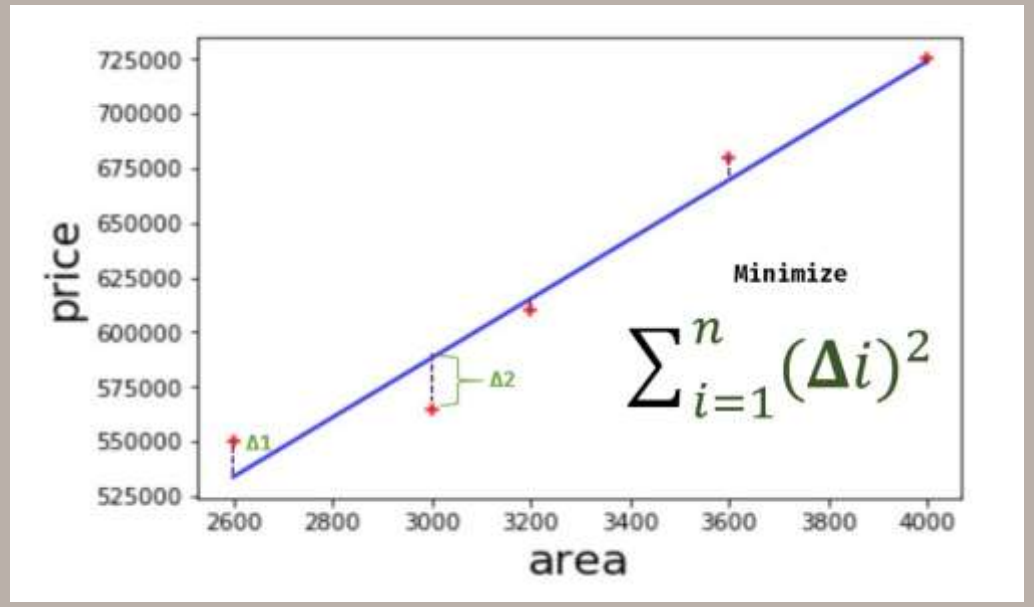
Week-02, c-2

You can draw multiple lines like this but we choose the one where total sum of error is minimum



To get the best line first draw a line and calculate the square sum of the distance of each point from the line which is called Mean Squared Error (MSE). Here

MSE= $\sum_{i=1}^{n}(\Delta i)^2$

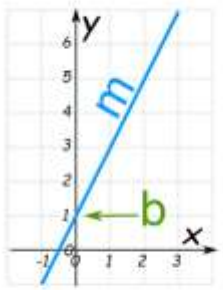


Calculate MSE for each line and select the line whose MSE is minimum. To make all the distance positive by squaring it.

Linear Regression: SML

Week-02, c-3

Home prices can be presented as following equation,
home **price = m * (area) + b**
Generic form of same equation is,



price = m * area + b

y = mx + b

Slope (or Gradient) Y Intercept

Reference: https://www.mathsisfun.com/algebra/linear-equations.html

Program Listing in Python

```
import pandas as pd
import numpy as np
from sklearn import linear_model
import matplotlib.pyplot as plt
```

```
df=pd.read_csv('F:\\python\\Houseprice.csv')
df
```

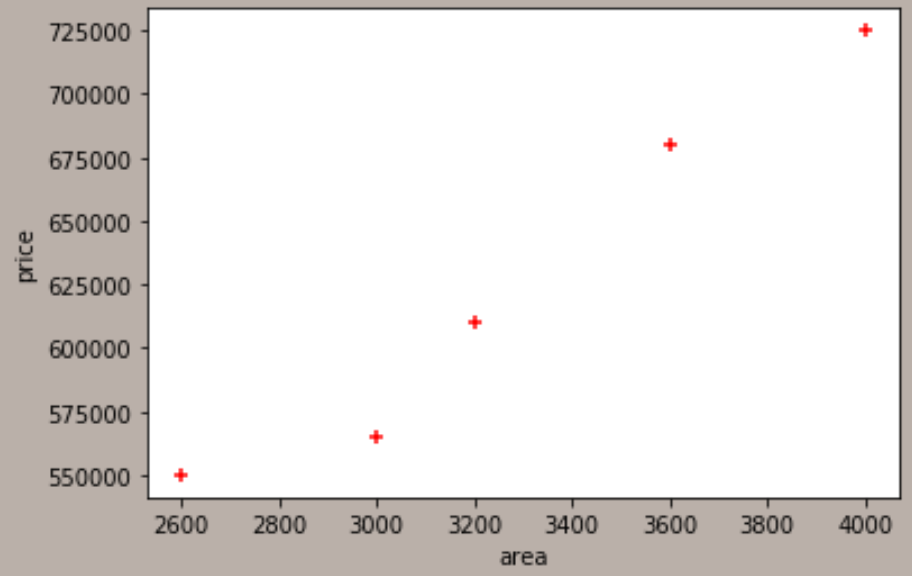
	area	price
0	2600	550000
1	3000	565000
2	3200	610000
3	3600	680000
4	4000	725000

Linear Regression: SML

Week-02, c-3

Program Listing in Python

```
%matplotlib inline
plt.xlabel('area')
plt.ylabel('price')
plt.scatter(df.area,df.price,color='red'
,marker='+')
```




```
new_df = df.drop('price',axis='columns')
new_df
```

	area
0	2600
1	3000
2	3200
3	3600
4	4000

```
price = df.price
price
```

0	550000
1	565000
2	610000
3	680000
4	725000

Linear Regression: SML

Week-02, c-3

Program Listing in Python

```
# Create linear regression object
reg = linear_model.LinearRegression()
reg.fit(new_df,price)
```

```
reg.predict([[3300]])
```

array([628715.75342466])

```
reg.coef_
```

array([135.78767123])

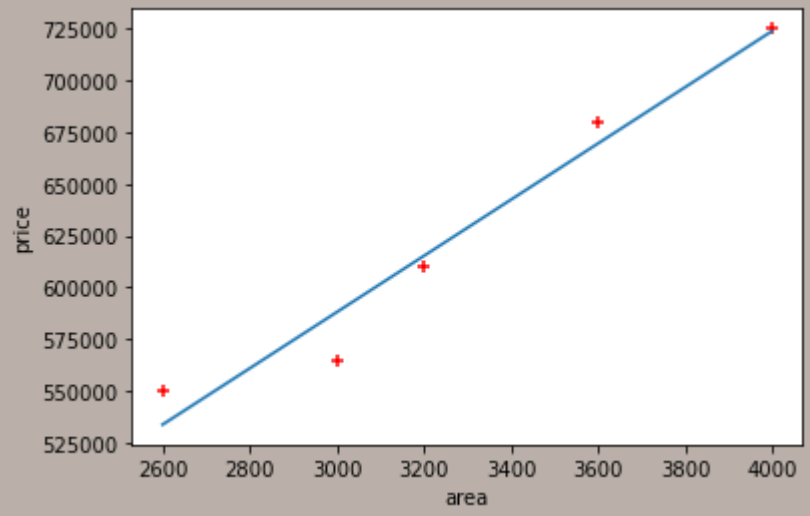
```
reg.intercept_
```

180616.43835616432

```
3300*135.78767123 + 180616.43835616432
```

628715.7534151643

```
%matplotlib inline
plt.xlabel('area')
plt.ylabel('price')
plt.scatter(df.area,df.price,color='red',marker='+')
plt.plot(area,reg.predict(area))
```



Linear Regression: SML

Week-02, c-3

Verification with Desmos

